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Selective vehicle slowdown

Abstract

A system for initiating an automated slowdown procedure within a vehicle includes a system controller adapted to detect a request for initiation of an automated slowdown procedure, a human machine interface (HMI), wherein the system controller is adapted to prompt, via the HMI, the driver of the vehicle for verification to initiate the automated slowdown procedure, and a vehicle control module, wherein the system controller is adapted to initiate, via the vehicle control module, the automated slowdown procedure when either one of the driver provides, via the HMI, verification to initiate the automated slowdown procedure, or a response from the driver is not received within a pre-determined time.

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Background/Summary

INTRODUCTION

(1) The present disclosure relates to a system and method for allowing a driver or passenger within a vehicle to initiate an automated vehicle slowdown when the driver or passenger determines that the driver is no longer able to properly operate the vehicle.

(2) Vehicles are equipped with many sensors to monitor the driver and to take measures in response to detecting that a driver of a vehicle is distracted or otherwise no longer able to properly operate the vehicle. However, current systems do not provide the ability for the driver or a passenger within the vehicle to, upon realizing that the driver is no longer able to properly operate the vehicle, initiate an automated vehicle slowdown.

(3) Thus, while current systems and methods achieve their intended purpose, there is a need for a new and improved system and method for providing the ability for the driver or a passenger within the vehicle to, upon realizing that the driver is no longer able to properly operate the vehicle, initiate an automated vehicle slowdown.

SUMMARY

(4) According to several aspects of the present disclosure, a method of initiating an automated slowdown procedure within a vehicle includes providing, with a system controller, control of the vehicle to a driver of the vehicle, wherein the driver of the vehicle proceeds with driving the vehicle, detecting, with the system controller, a request for initiation of an automated slowdown procedure, prompting, with the system controller via a human machine interface (HMI), the driver of the vehicle for verification to initiate the automated slowdown procedure, and initiating, with the system controller, via a vehicle control module, the automated slowdown procedure when either one of the driver provides, via the HMI, verification to initiate the automated slowdown procedure, or a response from the driver is not received within a pre-determined time.

(5) According to another aspect, the method further includes, prior to initiating the automated slowdown procedure, engaging an automated driving mode when the vehicle is not already in an automated driving mode.

(6) According to another aspect, the method further includes, after initiating the automated slowdown procedure, prompting, with the system controller, via the HMI, the driver of the vehicle with an option to selectively over-ride the automated slowdown procedure, and one of continuing with the automated slowdown procedure when the driver of the vehicle does not selectively over-ride, via the HMI, the automated slowdown procedure, or returning, with the system controller, control of the vehicle to the driver of the vehicle when the driver of the vehicle selectively over-rides, via the HMI, the automated slowdown procedure.

(7) According to another aspect, the method further includes, prior to the providing, with the system controller, control of the vehicle to the driver of the vehicle, and the driver of the vehicle proceeding with driving, receiving, with the system controller, via the HMI, preferences from the driver related to initiation of the automated slowdown procedure.

(8) According to another aspect, the method further includes, after receiving, with the system controller, via the HMI, preferences from the driver related to initiation of the automated slowdown procedure, detecting, with an occupant monitoring system, the presence and location of passengers, in addition to the driver, within the vehicle, identifying, with the occupant monitoring system, at least one passenger located in a rear seating area of the vehicle, and determining, with the system controller, based on preferences from the driver, eligibility of the at least one passenger located in the rear seating area of the vehicle to request initiation of the automated slowdown procedure.
(9) According to another aspect, the detecting, with the system controller, a request for initiation of an automated slowdown procedure further includes receiving, with the system controller, via a front seat actuator, a request for initiation of the slowdown procedure from at least one of the driver of the vehicle and a front seat passenger of the vehicle.
(10) According to another aspect, the detecting, with the system controller, a request for initiation of an automated slowdown procedure further includes receiving, with the system controller, via a rear seating area actuator, a request for initiation of the slowdown procedure from the at least one passenger located in the rear seating area when the system controller determines, based on preferences from the driver, that the at least one passenger located within the rear seating area is eligible to request initiation of the automated slowdown procedure.
(11) According to another aspect, the receiving, with the system controller, via the HMI, preferences from the driver related to initiation of the automated slowdown procedure further includes receiving preferences from the driver including at least one of disabling the rear seating area actuator, disabling both the rear seating area actuator and the front seat actuator, disabling one or both of the front seat actuator and the rear seating area actuator for a single ignition cycle, disabling the eligibility of a specific passenger to initiate a request for the vehicle slowdown procedure, and disabling the eligibility of any passenger that is less than a pre-set age.
(12) According to another aspect, each of the front seat actuator and the rear seating area actuator is at least one of a mechanical actuator, and an icon presented on an HMI.
(13) According to another aspect, the initiating, with the system controller, via the vehicle control module, the automated slowdown procedure further includes actuating, with the system controller, automated driving algorithms within the vehicle control module, disabling driver control of the vehicle, and causing, with the vehicle control module, the vehicle to slow down, maneuver off active lanes of a roadway on which the vehicle is traveling, and stop.
(14) According to several aspects of the present disclosure, a system for initiating an automated slowdown procedure within a vehicle includes a system controller adapted to detect a request for initiation of an automated slowdown procedure, a human machine interface (HMI), wherein the system controller is adapted to prompt, via the HMI, the driver of the vehicle for verification to initiate the automated slowdown procedure, and a vehicle control module, wherein the system controller is adapted to initiate, via the vehicle control module, the automated slowdown procedure when either one of the driver provides, via the HMI, verification to initiate the automated slowdown procedure, or a response from the driver is not received within a pre-determined time.
(15) According to another aspect, prior to initiating the automated slowdown procedure, the system controller is adapted to engage an automated driving mode when the vehicle is not already in an automated driving mode.
(16) According to another aspect, after initiating the automated slowdown procedure, the system controller is further adapted to prompt, via the HMI, the driver of the vehicle with an option to selectively over-ride the automated slowdown procedure, and one of continue with the automated slowdown procedure when the driver of the vehicle does not selectively over-ride, via the HMI, the automated slowdown procedure, or return control of the vehicle to the driver of the vehicle when the driver of the vehicle selectively over-rides, via the HMI, the automated slowdown procedure.
(17) According to another aspect, prior to detecting a request for initiation of an automated slowdown procedure, the system controller is adapted to receive, via the HMI, preferences from the

driver related to initiation of the automated slowdown procedure.
(18) According to another aspect, after receiving, via the HMI, preferences from the driver related to initiation of the automated slowdown procedure, the system controller is further adapted to detect, with an occupant monitoring system, the presence and location of passengers, in addition to the driver, within the vehicle, identify, with the occupant monitoring system, at least one passenger located in a rear seating area of the vehicle, and determine, based on preferences from the driver, eligibility of the at least one passenger located in the rear seating area of the vehicle to request initiation of the automated slowdown procedure.
(19) According to another aspect, when detecting a request for initiation of an automated slowdown procedure, the system controller is further adapted to at least one of receive, via a front seat actuator, a request for initiation of the slowdown procedure from at least one of the driver of the vehicle and a front seat passenger of the vehicle, and receive, via a rear seating area actuator, a request for initiation of the slowdown procedure from the at least one passenger located in the rear seating area when the system controller determines, based on preferences from the driver, that the at least one passenger located within the rear seating area is eligible to request initiation of the automated slowdown procedure.
(20) According to another aspect, the system controller, via the HMI, is adapted to receive preferences from the driver related to initiation of the automated slowdown procedure including at least one of disabling the rear seating area actuator, disabling both the rear seating area actuator and the front seat actuator, disabling one or both of the front seat actuator and the rear seating area actuator for a single ignition cycle, disabling the eligibility of a specific passenger to initiate a request for the vehicle slowdown procedure, and disabling the eligibility of any passenger that is less than a pre-set age.
(21) According to another aspect, each of the front seat actuator and the rear seating area actuator is at least one of a mechanical actuator, and an icon presented on an HMI.
(22) According to another aspect, when initiating, via the vehicle control module, the automated slowdown procedure, the system controller is further adapted to actuate automated driving algorithms within the vehicle control module, disable driver control of the vehicle, and cause, with the vehicle control module, the vehicle to slow down, maneuver off active lanes of a roadway on which the vehicle is traveling and stop.
(23) Further areas of applicability will become apparent from the description provided herein. It should be understood that the description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The drawings described herein are for illustration purposes only and are not intended to limit the scope of the present disclosure in any way.
(2) FIG. **1** is a schematic diagram of a vehicle having a system for providing a warning to an occupant of a hazardous object prior to the occupant exiting the vehicle according to an exemplary embodiment;
(3) FIG. **2** is a schematic diagram of the system according to an exemplary embodiment;
(4) FIG. **3** is a schematic top view of a vehicle including a front seat actuator and a rear seat actuator for requesting initiation of an automated slowdown procedure;
(5) FIG. **4** is a schematic diagram of a dashboard of a vehicle having a system in accordance with the present disclosure, a front seat actuator and a driver HMI;
(6) FIG. **5** is a schematic view of an HMI displaying an icon for either the front seat actuator or the rear seating area actuator according to an exemplary embodiment;

(7) FIG. **6** is a schematic view of an HMI displaying a prompt for verification by the driver to initiate an automated slowdown procedure;
(8) FIG. **7** is a schematic view of an HMI displaying an over-ride prompt for the driver;
(9) FIG. **8** is a flow chart illustrating a method of providing the ability for the driver or a passenger within the vehicle to initiate an automated vehicle slowdown procedure;
(10) FIG. **9** is a flow chart illustrating an exemplary embodiment of the detecting, with the system controller, if a request for initiation of an automated slowdown procedure has been received by the system controller at block **104** of FIG. **8**; and
(11) FIG. **10** is a flow chart illustrating an exemplary embodiment of the initiating, with the system controller, via the vehicle control module, the automated slowdown procedure at block **112** of FIG. **8**.
(12) The figures are not necessarily to scale and some features may be exaggerated or minimized, such as to show details of particular components. In some instances, well-known components, systems, materials or methods have not been described in detail in order to avoid obscuring the present disclosure. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the present disclosure.

DETAILED DESCRIPTION

(13) The following description is merely exemplary in nature and is not intended to limit the present disclosure, application, or uses. Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding technical field, background, brief summary or the following detailed description. It should be understood that throughout the drawings, corresponding reference numerals indicate like or corresponding parts and features. As used herein, the term module refers to any hardware, software, firmware, electronic control component, processing logic, and/or processor device, individually or in any combination, including without limitation: application specific integrated circuit (ASIC), an electronic circuit, a processor (shared, dedicated, or group) and memory that executes one or more software or firmware programs, a combinational logic circuit, and/or other suitable components that provide the described functionality. Although the figures shown herein depict an example with certain arrangements of elements, additional intervening elements, devices, features, or components may be present in actual embodiments. It should also be understood that the figures are merely illustrative and may not be drawn to scale.
(14) As used herein, the term “vehicle” is not limited to automobiles. While the present technology is described primarily herein in connection with automobiles, the technology is not limited to automobiles. The concepts can be used in a wide variety of applications, such as in connection with aircraft, marine craft, other vehicles, and consumer electronic components.
(15) In accordance with an exemplary embodiment, FIG. **1** shows a vehicle **10** with an associated system **11** for providing the ability for the driver or a passenger within the vehicle **10** to, upon realizing that the driver is no longer able to properly operate the vehicle **10**, initiate an automated vehicle slowdown procedure in accordance with various embodiments. In general, the system **11** works in conjunction with other systems within the vehicle **10** to display various information and infotainment content for the passenger. The vehicle **10** generally includes a chassis **12**, a body **14**, front wheels **16**, and rear wheels **18**. The body **14** is arranged on the chassis **12** and substantially encloses components of the vehicle **10**. The body **14** and the chassis **12** may jointly form a frame. The front wheels **16** and rear wheels **18** are each rotationally coupled to the chassis **12** near a respective corner of the body **14**.
(16) In various embodiments, the vehicle **10** is an autonomous vehicle and the system **11** is incorporated into the autonomous vehicle **10**. An autonomous vehicle **10** is, for example, a vehicle **10** that is automatically controlled to carry passengers from one location to another. The vehicle **10** is depicted in the illustrated embodiment as a passenger car, but it should be appreciated that any

other vehicle including motorcycles, trucks, sport utility vehicles (SUVs), recreational vehicles (RVs), etc., can also be used. In an exemplary embodiment, the vehicle **10** is equipped with a so-called Level Four or Level Five automation system. A Level Four system indicates “high automation”, referring to the driving mode-specific performance by an automated driving system of all aspects of the dynamic driving task, even if a human driver does not respond appropriately to a request to intervene. A Level Five system indicates “full automation”, referring to the full-time performance by an automated driving system of all aspects of the dynamic driving task under all roadway and environmental conditions that can be managed by a human driver. The novel aspects of the present disclosure are also applicable to non-autonomous vehicles.

(17) As shown, the vehicle **10** generally includes a propulsion system **20**, a transmission system **22**, a steering system **24**, a brake system **26**, a sensor system **28**, an actuator system **30**, at least one data storage device **32**, a vehicle controller **34**, and a wireless communication module **36**. In an embodiment in which the vehicle **10** is an electric vehicle, there may be no transmission system **22**. The propulsion system **20** may, in various embodiments, include an internal combustion engine, an electric machine such as a traction motor, and/or a fuel cell propulsion system. The transmission system **22** is configured to transmit power from the propulsion system **20** to the vehicle's front wheels **16** and rear wheels **18** according to selectable speed ratios. According to various embodiments, the transmission system **22** may include a step-ratio automatic transmission, a continuously-variable transmission, or other appropriate transmission. The brake system **26** is configured to provide braking torque to the vehicle's front wheels **16** and rear wheels **18**. The brake system **26** may, in various embodiments, include friction brakes, brake by wire, a regenerative braking system such as an electric machine, and/or other appropriate braking systems. The steering system **24** influences a position of the front wheels **16** and rear wheels **18**. While depicted as including a steering wheel for illustrative purposes, in some embodiments contemplated within the scope of the present disclosure, the steering system **24** may not include a steering wheel.

(18) The sensor system **28** includes one or more sensing devices **40***a*-**40***n* that sense observable conditions of the exterior environment and/or the interior environment of the autonomous vehicle **10**. The sensing devices **40***a*-**40***n* can include, but are not limited to, radars, lidars, global positioning systems, optical cameras, thermal cameras, ultrasonic sensors, and/or other sensors. The cameras can include two or more digital cameras spaced at a selected distance from each other, in which the two or more digital cameras are used to obtain stereoscopic images of the surrounding environment in order to obtain a three-dimensional image or map. The plurality of sensing devices **40***a*-**40***n* is used to determine information about an environment surrounding the vehicle **10**. In an exemplary embodiment, the plurality of sensing devices **40***a*-**40***n* includes at least one of a motor speed sensor, a motor torque sensor, an electric drive motor voltage and/or current sensor, an accelerator pedal position sensor, a coolant temperature sensor, a cooling fan speed sensor, and a transmission oil temperature sensor. In another exemplary embodiment, the plurality of sensing devices **40***a*-**40***n* further includes sensors to determine information about the environment surrounding the vehicle **10**, for example, an ambient air temperature sensor, a barometric pressure sensor, and/or a photo and/or video camera which is positioned to view the environment in front of the vehicle **10**. In another exemplary embodiment, at least one of the plurality of sensing devices **40***a*-**40***n* is capable of measuring distances in the environment surrounding the vehicle **10**.

(19) In a non-limiting example wherein the plurality of sensing devices **40***a*-**40***n* includes a camera, the plurality of sensing devices **40***a*-**40***n* measures distances using an image processing algorithm configured to process images from the camera and determine distances between objects. In another non-limiting example, the plurality of vehicle sensors **40***a*-**40***n* includes a stereoscopic camera having distance measurement capabilities. In one example, at least one of the plurality of sensing devices **40***a*-**40***n* is affixed inside of the vehicle **10**, for example, in a headliner of the vehicle **10**, having a view through the windshield of the vehicle **10**. In another example, at least one of the plurality of sensing devices **40***a*-**40***n* is a camera affixed outside of the vehicle **10**, for example, on

a roof of the vehicle **10**, having a view of the environment surrounding the vehicle **10** and adapted to collect information (images) related to the environment outside the vehicle **10**. It should be understood that various additional types of sensing devices, such as, for example, LiDAR sensors, ultrasonic ranging sensors, radar sensors, and/or time-of-flight sensors are within the scope of the present disclosure. The actuator system **30** includes one or more actuator devices **42***a*-**42***n* that control one or more vehicle **10** features such as, but not limited to, the propulsion system **20**, the transmission system **22**, the steering system **24**, and the brake system **26**.
(20) The vehicle controller **34** includes at least one processor **44** and a computer readable storage device or media **46**. The at least one data processor **44** can be any custom made or commercially available processor, a central processing unit (CPU), a graphics processing unit (GPU), an auxiliary processor among several processors associated with the vehicle controller **34**, a semi-conductor based microprocessor (in the form of a microchip or chip set), a macro-processor, any combination thereof, or generally any device for executing instructions. The computer readable storage device or media **46** may include volatile and nonvolatile storage in read-only memory (ROM), random-access memory (RAM), and keep-alive memory (KAM), for example. KAM is a persistent or non-volatile memory that may be used to store various operating variables while the at least one data processor **44** is powered down. The computer-readable storage device or media **46** may be implemented using any of a number of known memory devices such as PROMs (programmable read-only memory), EPROMs (electrically PROM), EEPROMs (electrically erasable PROM), flash memory, or any other electric, magnetic, optical, or combination memory devices capable of storing data, some of which represent executable instructions, used by the controller **34** in controlling the vehicle **10**.
(21) The instructions may include one or more separate programs, each of which includes an ordered listing of executable instructions for implementing logical functions. The instructions, when executed by the at least one processor **44**, receive and process signals from the sensor system **28**, perform logic, calculations, methods and/or algorithms for automatically controlling the components of the vehicle **10**, and generate control signals to the actuator system **30** to automatically control the components of the vehicle **10** based on the logic, calculations, methods, and/or algorithms. Although only one controller **34** is shown in FIG. **1**, embodiments of the vehicle **10** can include any number of controllers **34** that communicate over any suitable communication medium or a combination of communication mediums and that cooperate to process the sensor signals, perform logic, calculations, methods, and/or algorithms, and generate control signals to automatically control features of the autonomous vehicle **10**.
(22) In various embodiments, one or more instructions of the vehicle controller **34** are embodied in a trajectory planning system and, when executed by the at least one data processor **44**, generates a trajectory output that addresses kinematic and dynamic constraints of the environment. For example, the instructions receive as input process sensor and map data. The instructions perform a graph-based approach with a customized cost function to handle different road scenarios in both urban and highway roads.
(23) The wireless communication module **36** is configured to wirelessly communicate information to and from other remote entities **48**, such as but not limited to, other vehicles (“V2V” communication,) infrastructure (“V2I” communication), remote systems, remote servers, cloud computers, and/or personal devices. In an exemplary embodiment, the communication system **36** is a wireless communication system configured to communicate via a wireless local area network (WLAN) using IEEE 802.11 standards or by using cellular data communication. However, additional or alternate communication methods, such as a dedicated short-range communications (DSRC) channel, are also considered within the scope of the present disclosure. DSRC channels refer to one-way or two-way short-range to medium-range wireless communication channels specifically designed for automotive use and a corresponding set of protocols and standards.
(24) The vehicle controller **34** is a non-generalized, electronic control device having a

preprogrammed digital computer or processor, memory or non-transitory computer readable medium used to store data such as control logic, software applications, instructions, computer code, data, lookup tables, etc., and a transceiver [or input/output ports]. Computer readable medium includes any type of medium capable of being accessed by a computer, such as read only memory (ROM), random access memory (RAM), a hard disk drive, a compact disc (CD), a digital video disc (DVD), or any other type of memory. A "non-transitory" computer readable medium excludes wired, wireless, optical, or other communication links that transport transitory electrical or other signals. A non-transitory computer readable medium includes media where data can be permanently stored and media where data can be stored and later overwritten, such as a rewritable optical disc or an erasable memory device. Computer code includes any type of program code, including source code, object code, and executable code.

(25) Referring to FIG. **2** a schematic diagram of the system **11** is shown. The system **11** includes a system controller **34**A in communication with the plurality of sensing devices (onboard sensors) **40***a*-**40***n*. The system controller **34**A may be the vehicle controller **34**, or the system controller **34**A may be a separate controller in communication with the vehicle controller **34**. In addition to the plurality of onboard sensors **40***a*-**40***n*, the system controller **34**A is in communication with an occupant monitoring system **50**, a vehicle control module **52**, a front seat actuator **54**, adapted to allow a driver **56** or a front seat passenger **58**A to provide input to the system **11** to request an automated vehicle slowdown procedure, a rear seating area actuator **60** adapted to allow a passenger **58***b* seated in a rear seating area **62** of the vehicle **10** to provide input to the system **11** to request an automated vehicle slowdown procedure and a human machine interface (HMI) **64**A having a touch screen display adapted to allow interaction between the driver **56** and the system **11**.

(26) Referring to FIG. **3**, an example of a vehicle **10** having a system **11** in accordance with the present disclosure is shown, wherein the vehicle **10** includes a driver seat **66** for the driver **56** of the vehicle **10**, a front passenger seat **68** for the front passenger **58**A of the vehicle **10**, and two rear passenger seats **70** for two rear passengers **58**B seated in the rear seating area **62** of the vehicle **10**. In an exemplary embodiment, the vehicle **10** includes a front seat actuator **54** that is a mechanical button or knob **54**A located centrally on the dashboard, armrest or headliner (as shown within FIG. **3** and FIG. **4**, centrally on the dashboard) within the vehicle **10**, wherein the front seat actuator **54**, **54**A is accessible by either the driver **56** or the front seat passenger **58**A. The vehicle **10** further includes a rear seating area actuator **60** that is a mechanical button or knob **60**A located centrally within the rear seating area **62**, wherein the rear seating area actuator **60**, **60**A is accessible by any of the rear seat passengers **58**B. The system controller **34**A is adapted to receive, via the front seat actuator **54**, a request for initiation of the slowdown procedure from at least one of the driver **56** of the vehicle **10** and the front seat passenger **58**A of the vehicle **10**, and to receive, via the rear seating area actuator **60**, a request for initiation of the slowdown procedure from the at least one rear seat passenger **58**B located in the rear seating area **62**.

(27) Referring to FIG. **5**, in another exemplary embodiment, the front seat actuator **54** is an icon **54**B or "soft button" displayed on the driver HMI **64**A located centrally on the dashboard, wherein the front seat actuator **54**, **54**B is accessible by either the driver **56** or the front seat passenger **58**A. Alternatively, the front seat actuator **54** includes the icon **54**B displayed on a touch screen **72**A of the driver HMI **64**A, as discussed above, to allow the driver **56** to request an automated vehicle slowdown procedure, and the front seat actuator **54** further includes an icon **54**C displayed on a touch screen **72**B of a front seat passenger HMI **64**B, to allow the front seat passenger **58**A to request an automated vehicle slowdown procedure. The rear seating area **62** includes a rear seat passenger HMI **64**C for each one of the rear seat passengers **58**B, wherein the rear seating area actuator **60** includes an icon **60**B displayed on a touch screen **72**C of each one of the rear seat passenger HMIs **64**C to allow each of the rear seat passengers **58**B to request initiation of an automated vehicle slowdown procedure. It should be understood that FIG. **5** represents a schematic illustration representing each of the driver HMI **64**A, the front seat passenger HMI **64**B and each of

the rear seat passenger HMIs **64**C, wherein the displayed icons **54**B, **54**C, **60**B for the front seat actuator **54** and the rear seating area actuator **60** are the same. It should be understood that the displayed icons **54**B, **54**C, **60**B could be different without departing from the scope of the present disclosure.

(28) In still another exemplary embodiment, each of the driver HMI **64**A, the front seat passenger HMI **64**B and the rear seat passenger HMIs **64**C include a microphone **74** adapted to allow the driver **56**, the front seat passenger **58**A and the rear seat passengers **58**B to provide verbal input to the system **11**, and the microphone **74** acts as the front seat actuator **54** and rear seating area actuator **60** allowing the driver **56**, front seat passenger **58**A or rear passengers **58**B to verbally request initiation of an automated vehicle slowdown procedure.

(29) The system controller **34**A detects when a request for an automated slowdown procedure has been requested by the driver **56** or the front seat passenger **58**A, by actuating the front seat actuator **54**, or by a rear seating area passenger **58**B, by actuating the rear seating area actuator **60**. When the system controller **34**A detects that a request for an automated slowdown procedure has been requested, the system controller **34**A, prompts the driver **56**, via the driver HMI **64**A, for verification to initiate the automated slowdown procedure. Referring to FIG. **6**, in an exemplary embodiment, the system controller **34**A displays a prompt **76** asking the driver **56** “Automated Vehicle Slowdown?”, wherein the driver can respond to the prompt by selecting, by touching the touchscreen display **72**A, either “YES”, “NO” or “DISABLE”. This provides an opportunity for the driver **56**, upon seeing the prompt **76**, to either verify, by selecting “YES”, or deny, by selecting “NO”, that the driver **56** wants to allow initiation of an automated slowdown procedure, in the event that the driver **56** requested the automated slowdown procedure, and to either verify, by selecting “YES”, or deny, by selecting “NO”, that the driver **56** wants to allow initiation of an automated slowdown procedure, in the even that the automated slowdown procedure was requested by the front seat passenger **58**A or a rear seat passenger **58**B. Further, the displayed prompt **76** provides the ability for the driver, upon seeing the prompt **76**, to disable the system **11**, denying the current request for an automated slowdown procedure and disabling the ability for further requests by selecting “DISABLE”. Thus, if the driver **56** inadvertently actuates the front seat actuator **54**, the displayed prompt **76** allows the driver **56** to cancel the request before the system controller **34**A initiates the automated slowdown procedure. Further, if the driver **56** sees that a request has been made by another passenger **58**A, **58**B, the driver may realize that they are no longer able to properly operate the vehicle **10**, and allow the automated slowdown procedure, or the driver **56** may not feel that the request was warranted and deny the request.

(30) When the system controller **34**A displays the prompt **76**, if the system controller **34**A does not receive, via the driver HMI **64**A, a response from the driver **56** within a predetermined time frame, the system controller **34**A will take the lack of response as acceptance of the request. Thus, the system controller **34**A will proceed with initiation of an automated slowdown procedure upon the occurrence of either the driver **56** providing, via the driver HMI **64**A, verification of acceptance to initiate the automated slowdown procedure, or when a response from the driver **56** is not received within a pre-determined time. For example, a rear seat passenger **58**B requests an automated slowdown procedure and the system controller **34**A displays, via the driver HMI **64**A, a prompt **76** for verification by the driver **56**. If the driver **56** does not see and respond to the prompt **76** within the predetermined time frame, for example 10 seconds, then the system controller **34**A will proceed with initiation of the automated slowdown procedure.

(31) Initiation of an automated slowdown procedure requires the system controller **34**A to take control of the vehicle **10** away from the driver **56**, and to operate the vehicle **10** autonomously. In an exemplary embodiment, the system controller **34**A, prior to initiating the automated slowdown procedure, is adapted to engage an automated driving mode when the vehicle **10** is not already in an automated driving mode. When initiating, via the vehicle control module **52**, the automated slowdown procedure, the system controller **34**A is further adapted to actuate automated driving

algorithms within the vehicle control module **52**, disable driver control of the vehicle **10**, and cause, with the vehicle control module **52**, the vehicle **10** to slow down, maneuver off active lanes of a roadway on which the vehicle **10** is traveling and stop. Upon initiation of an automated slowdown procedure, the vehicle control module **52** is adapted to quickly and safely maneuver the vehicle **10** to a safe location by immediately causing the vehicle **10** to use automated driving algorithms such as, but not limited to adaptive cruise control (ACC) and lane keeping assistance (LKA) algorithms to maneuver the vehicle **10**, for example, to the furthest right lane of a highway, and eventually, off to the shoulder of the highway and then bringing the vehicle **10** to a stop.

(32) In an exemplary embodiment, after initiating the automated slowdown procedure, the system controller **34**A is further adapted to prompt, via the driver HMI **64**A, the driver **56** of the vehicle **10** with an option to selectively over-ride the automated slowdown procedure. Referring to FIG. **7**, for example, the system controller **34**A displays an over-ride prompt **78** "Discontinue Automated Vehicle Slowdown", wherein, the driver **56** may selectively, by touching the prompt **78** on the touch screen display **72**A of the driver HMI **64**A, over-ride the automated slowdown procedure, wherein the system controller **34**A will stop the automated slowdown procedure and return control of the vehicle **10** to the driver **56**. As long as the driver **56** does not selectively over-ride the automated slowdown procedure, the system controller **34**A will continue with the automated slowdown procedure.

(33) In an exemplary embodiment, prior to detecting a request for initiation of an automated slowdown procedure, the system controller **34**A is adapted to receive, via the driver HMI **64**A, preferences from the driver **56** related to initiation of the automated slowdown procedure. Thus, the driver **56**, can customize the system **11** as to when and how to allow initiation of automated slowdown procedures.

(34) For example, the driver **56** may be traveling with young children seated in the rear seating area **62**, and does not want them to be able to request initiation of an automated slowdown procedure. The driver may provide preferences that include disabling the rear seating area actuator **60**. Thus, the young children seated within the rear seating area **62** of the vehicle **10** will not be able to request an automated vehicle slowdown procedure.

(35) After receiving, via the driver HMI **64**A, preferences from the driver **56** related to initiation of the automated slowdown procedure, the system controller **34**A is further adapted to detect, with the occupant monitoring system **50**, the presence and location of passengers **58**A, **58**B, in addition to the driver **56**, within the vehicle **10**, identify, with the occupant monitoring system **50**, at least one rear seat passenger **58**B located in the rear seating area **62** of the vehicle **10**, and determine, based on preferences from the driver **56**, eligibility of the at least one rear seat passenger **58**B located in the rear seating area **62** of the vehicle **10** to request initiation of the automated slowdown procedure.

(36) The occupant monitoring system **50** includes sensors known in the art to capture images of the rear seat passengers **58**B and, using computer vision algorithms, identify the rear seat passengers **58**B. Identification of the rear seat passengers **58**B may include using computer vision algorithms to approximate an age of such rear seat passengers **58**B, or by using computer vision algorithms in conjunction with machine learning algorithms and data of previous drivers **56** and passengers **58**A, **58**B, determining a specific identity of such rear seat passengers **58**B.

(37) Thus, in an exemplary embodiment, the driver **56** may provide preferences that include disabling the rear seating area actuator **60** for rear seat passengers **58**B that are less than a certain age, for example, ten years old, wherein, the system controller **34**A, using the occupant monitoring system **50**, determines that the age of the rear seat passengers **58**B is approximated to be less than ten years old and disables the rear seating area actuator **60**. In another exemplary embodiment, the driver **56** may provide preferences that include disabling the rear seating area actuator **60** when the two young children (specifically, Johnny and Susie) are seated in the rear seating area **62**. The system controller **34**A, using the occupant monitoring system **50**, identifies Johnny and Susie

sitting in the rear seating area **62** and disables the rear seating area actuator **60**.
(38) Such preferences may be applied permanently, wherein the rear seating area actuator **60** is disabled until the driver **56** changes the preferences, or such preferences may apply only to the current ignition cycle of the vehicle **10**, wherein the next time the vehicle **10** is used, such preferences no longer apply, and the system **11** will operate on default settings.
(39) The system controller **34**A is adapted to receive, via the front seat actuator **54**, a request for initiation of the slowdown procedure from at least one of the driver **56** of the vehicle **10** and a front seat passenger **58**A of the vehicle **10**, and to receive, via the rear seating area actuator **60**, a request for initiation of the slowdown procedure from the at least one rear seat passenger **58**B located in the rear seating area **62** when the system controller **34**A determines, based on preferences from the driver **56**, that the at least one rear seat passenger **58**B is eligible to request initiation of the automated slowdown procedure. Thus, in an exemplary embodiment, the driver **56** may provide preferences that include disabling the rear seating area actuator **60** for rear seat passengers **58**B that are less than a certain age, for example, ten years old, wherein, the system controller **34**A, using the occupant monitoring system **50**, determines that the age of the rear seat passengers **58**B is approximated to be more than ten years old and enables use of the rear seating area actuator **60** by such rear seat passengers **58**B to request an automated slowdown procedure.
(40) In various embodiments, the system controller **34**A, via the driver HMI **64**A, is adapted to receive preferences from the driver **56** related to initiation of the automated slowdown procedure including, but not limited to, at least one of disabling the rear seating area actuator **60**, disabling both the rear seating area actuator **60** and the front seat actuator **54**, disabling one or both of the front seat actuator **54** and the rear seating area actuator **60** for a single ignition cycle, or until the driver **56** changes the preferences, disabling the eligibility of a specific passenger, as identified by the occupant monitoring system **50**, to initiate a request for the vehicle slowdown procedure, and disabling the eligibility of any passenger **58**A, **58**B that is less than a pre-set age.
(41) Referring to FIG. **8** a method **100** of initiating an automated slowdown procedure within a vehicle **10** includes, beginning at block **102**, providing, with a system controller **34**A, control of the vehicle **10** to a driver **56** of the vehicle **10**, wherein the driver **56** of the vehicle **10** proceeds with driving the vehicle **10**, moving to block **104**, detecting, with the system controller **34**A, a request for initiation of an automated slowdown procedure, wherein, if, at block **104**, the system controller does not detect a request for initiation of an automated slowdown procedure, then, moving from block **104** to block **106**, no action is taken, and moving from block **106** to **102**, control of the vehicle **10** remains with the driver **56**, and if, at block **104**, the system controller does detect a request for initiation of an automated slowdown procedure, then, moving from block **104** to block **108**, the method **100** includes prompting, with the system controller **34**A via a human machine interface (HMI) **64**A, the driver **56** of the vehicle **10** for verification to initiate the automated slowdown procedure, and, moving to block **112**, initiating, with the system controller **34**A, via a vehicle control module **52**, the automated slowdown procedure when, at block **110**, the system controller determines that either one of 1) the driver **56** provides, via the HMI **64**A, verification to initiate the automated slowdown procedure, or 2) a response from the driver **56** is not received within a pre-determined time.
(42) In an exemplary embodiment, the method **100** further includes, at block **114**, prior to initiating the automated slowdown procedure at block **112**, engaging an automated driving mode when the vehicle **10** is not already in an automated driving mode.
(43) In an exemplary embodiment, the method **100** further includes, after initiating the automated slowdown procedure at block **112**, moving to block **116**, prompting, with the system controller **34**A, via the HMI **64**A, the driver **56** of the vehicle **10** with an option to selectively over-ride the automated slowdown procedure, and moving to block **118**, one of, moving from block **118** to block **112**, continuing with the automated slowdown procedure when the driver **56** of the vehicle **10** does not selectively over-ride, via the HMI **64**A, the automated slowdown procedure, or, moving from

block **118** to block **120**, returning, with the system controller **34**A, control of the vehicle **10** to the driver **56** of the vehicle **10** when the driver **56** of the vehicle selectively over-rides, via the HMI **64**A, the automated slowdown procedure.

(44) In an exemplary embodiment, the method **100** further includes, prior to the providing, with the system controller **34**A, control of the vehicle **10** to the driver **56** of the vehicle **10**, and the driver **56** of the vehicle **10** proceeding with driving at block **102**, moving to block **122**, receiving, with the system controller **34**A, via the HMI **64**A, preferences from the driver **56** related to initiation of the automated slowdown procedure.

(45) In an exemplary embodiment, the method **100** further includes, after receiving, with the system controller **34**A, via the HMI **64**A, preferences from the driver **56** related to initiation of the automated slowdown procedure at block **122**, moving to block **124**, detecting, with an occupant monitoring system **50**, the presence and location of passengers **58**A, **58**B, in addition to the driver **56**, within the vehicle **10**, moving to block **126**, identifying, with the occupant monitoring system **50**, at least one passenger **58**B located in a rear seating area **62** of the vehicle **10**, and, moving to block **128**, determining, with the system controller **34**A, based on preferences from the driver **56**, eligibility of the at least one passenger **58**B located in the rear seating area **62** of the vehicle **10** to request initiation of the automated slowdown procedure.

(46) In an exemplary embodiment, the detecting, with the system controller **34**A, if a request for initiation of an automated slowdown procedure has been received by the system controller **34**A at block **104** further includes, moving to block **130**, detecting, with the system controller **34**A, if the system **11** is in an active mode. If the system **11** is not active, then, moving from block **130** to block **106**, no action is taken. If the system **11** is active, then the method further includes, moving from block **130** to block **132**, receiving, with the system controller **34**A, via the front seat actuator **54**, a request for initiation of the slowdown procedure from at least one of the driver **56** of the vehicle **10** and the front seat passenger **58**A of the vehicle **10**. If at block **132**, the system controller **34**A receives, via the front seat actuator **54**, a request for initiation of the slowdown procedure from at least one of the driver **56** of the vehicle **10** and the front seat passenger **58**A of the vehicle **10**, then the method progresses to block **108**, where the driver **56** is prompted, with the system controller **34**A via the human machine interface (HMI) **64**A, for verification to initiate the automated slowdown procedure. If at block **132**, the system controller **34**A does not receive a request via the front seat actuator **54**, then the method **100** further includes, moving from block **132** to block **134**, receiving, with the system controller **34**A, via a rear seating area actuator **60**, a request for initiation of the slowdown procedure from the at least one passenger **58**B located in the rear seating area **62**. If, at block **134**, no request is received from the rear seating area actuator **60**, then, moving from block **134** to block **106**, no action is taken. If at block **134**, a request is received from the rear seating area actuator **60**, then, moving from block **134** to block **136**, the system controller **34**A determines, based on preferences from the driver, eligibility of the at least one passenger **58**B located within the rear seating area **60** to request initiation of the automated slowdown procedure. If at block **136**, the system controller **34**A determines, based on preferences of the driver **56**, that the rear seat passenger **58**B is not eligible to request an automated slowdown procedure, then, moving from block **136** to block **106**, no action is taken. If at block **136**, the system controller **34**A determines, based on preferences of the driver **56**, that the rear seat passenger **58**B is eligible to request an automated slowdown procedure, then the method **100** proceeds to block **108**, where the driver **56** is prompted, with the system controller **34**A via the human machine interface (HMI) **64**A, for verification to initiate the automated slowdown procedure.

(47) In an exemplary embodiment, the initiating, with the system controller **34**A, via the vehicle control module **52**, the automated slowdown procedure at block **112**, further includes, moving to block **138**, actuating, with the system controller **34**A, automated driving algorithms within the vehicle control module **52**, moving to block **140**, disabling driver control of the vehicle **10**, and, moving to block **142**, causing, with the vehicle control module **52**, the vehicle **10** to slow down,

maneuver off active lanes of a roadway on which the vehicle **10** is traveling, and stop.

(48) A system and method of the present disclosure offers the advantage of allowing a driver **56** or passengers **58**A, **58**B within a vehicle **10** to request initiation of an automated slowdown procedure to autonomously maneuver the vehicle **10** off the roadway and bring the vehicle **10** to a stop.

(49) The description of the present disclosure is merely exemplary in nature and variations that do not depart from the gist of the present disclosure are intended to be within the scope of the present disclosure. Such variations are not to be regarded as a departure from the spirit and scope of the present disclosure.

Claims

1. A method of initiating an automated slowdown procedure within a vehicle, comprising: receiving, with the system controller, via a driver human machine interface (HMI), preferences from the driver related to initiation of the automated slowdown procedure; detecting, with an occupant monitoring system, the presence and location of passengers, in addition to the driver, within the vehicle; identifying, with the occupant monitoring system, at least one passenger located in a rear seating area of the vehicle; determining, with the system controller, based on the preferences from the driver, eligibility of the at least one passenger located in the rear seating area of the vehicle to request initiation of the automated slowdown procedure; providing, with a system controller, control of the vehicle to a driver of the vehicle, wherein the driver of the vehicle proceeds with driving the vehicle; detecting, with the system controller, a request for initiation of the automated slowdown procedure; prompting, with the system controller via the driver HMI, the driver of the vehicle for verification to initiate the automated slowdown procedure; and initiating, with the system controller, via a vehicle control module, the automated slowdown procedure when either one of: the driver provides, via the driver HMI, verification to initiate the automated slowdown procedure; or a response from the driver is not received within a pre-determined time.
2. The method of claim 1, further including, prior to initiating the automated slowdown procedure, engaging an automated driving mode when the vehicle is not already in an automated driving mode.
3. The method of claim 2, further including, after initiating the automated slowdown procedure: prompting, with the system controller, via the driver HMI, the driver of the vehicle with an option to selectively over-ride the automated slowdown procedure, and one of: continuing with the automated slowdown procedure when the driver of the vehicle does not selectively over-ride, via the driver HMI, the automated slowdown procedure; or returning, with the system controller, control of the vehicle to the driver of the vehicle when the driver of the vehicle selectively over-rides, via the driver HMI, the automated slowdown procedure.
4. The method of claim 3, wherein the detecting, with the system controller, a request for initiation of an automated slowdown procedure further includes receiving, with the system controller, via a front seat actuator, a request for initiation of the slowdown procedure from at least one of the driver of the vehicle and a front seat passenger of the vehicle.
5. The method of claim 4, wherein the detecting, with the system controller, a request for initiation of an automated slowdown procedure further includes receiving, with the system controller, via a rear seating area actuator, a request for initiation of the slowdown procedure from the at least one passenger located in the rear seating area when the system controller determines, based on the preferences from the driver, that the at least one passenger located within the rear seating area is eligible to request initiation of the automated slowdown procedure.
6. The method of claim 5, wherein the receiving, with the system controller, via the driver HMI, the preferences from the driver related to initiation of the automated slowdown procedure further includes receiving preferences from the driver including at least one of: disabling the rear seating area actuator; disabling both the rear seating area actuator and the front seat actuator; disabling one

or both of the front seat actuator and the rear seating area actuator for a single ignition cycle; disabling the eligibility of a specific passenger to initiate a request for the vehicle slowdown procedure; and disabling the eligibility of any passenger that is less than a pre-set age.
7. The method of claim 6, wherein the front seat actuator is at least one of: a mechanical actuator; and an icon presented on either one of the driver HMI and a front seat passenger HMI.
8. The method of claim 7, wherein the rear seating area actuator is at least one of: a mechanical actuator; and an icon presented on a rear seat passenger HMI.
9. The method of claim 8, wherein each of the driver HMI, the front seat passenger HMI and the rear seat passenger HMI includes a microphone adapted to receive verbal input to the system controller, the detecting, with the system controller, a request for initiation of an automated slowdown procedure further including receiving, with the system controller, a verbal request for initiation of the automated slowdown procedure via the microphone within one of the driver HMI, the front seat passenger HMI and the rear seat passenger HMI.
10. The method of claim 1, wherein the initiating, with the system controller, via the vehicle control module, the automated slowdown procedure further includes: actuating, with the system controller, automated driving algorithms within the vehicle control module; disabling driver control of the vehicle; and causing, with the vehicle control module, the vehicle to slow down, maneuver off active lanes of a roadway on which the vehicle is traveling, and stop.
11. A system for initiating an automated slowdown procedure within a vehicle, comprising: a system controller adapted to: receive, via a driver human machine interface (HMI), preferences from the driver related to initiation of the automated slowdown procedure and detect a request for initiation of the automated slowdown procedure; detect, with an occupant monitoring system, the presence and location of passengers, in addition to the driver, within the vehicle; identify, with the occupant monitoring system, at least one passenger located in a rear seating area of the vehicle; determine, based on the preferences from the driver, eligibility of the at least one passenger located in the rear seating area of the vehicle to request initiation of the automated slowdown procedure; and prompt, via the driver HMI, the driver of the vehicle for verification to initiate the automated slowdown procedure; and a vehicle control module, wherein the system controller is adapted to initiate, via the vehicle control module, the automated slowdown procedure when either one of: the driver provides, via the driver HMI, verification to initiate the automated slowdown procedure; or a response from the driver is not received within a pre-determined time.
12. The system of claim 11, wherein, prior to initiating the automated slowdown procedure, the system controller is adapted to engage an automated driving mode when the vehicle is not already in an automated driving mode.
13. The system of claim 12, wherein, after initiating the automated slowdown procedure, the system controller is further adapted to: prompt, via the driver HMI, the driver of the vehicle with an option to selectively over-ride the automated slowdown procedure, and one of: continue with the automated slowdown procedure when the driver of the vehicle does not selectively over-ride, via the driver HMI, the automated slowdown procedure; or return control of the vehicle to the driver of the vehicle when the driver of the vehicle selectively over-rides, via the driver HMI, the automated slowdown procedure.
14. The system of claim 13, wherein when detecting a request for initiation of an automated slowdown procedure, the system controller is further adapted to at least one of: receive, via a front seat actuator, a request for initiation of the slowdown procedure from at least one of the driver of the vehicle and a front seat passenger of the vehicle; and receive, via a rear seating area actuator, a request for initiation of the slowdown procedure from the at least one passenger located in the rear seating area when the system controller determines, based on the preferences from the driver, that the at least one passenger located within the rear seating area is eligible to request initiation of the automated slowdown procedure.
15. The system of claim 14, wherein the system controller, via the driver HMI, is adapted to receive

the preferences from the driver related to initiation of the automated slowdown procedure including at least one of: disabling the rear seating area actuator; disabling both the rear seating area actuator and the front seat actuator; disabling one or both of the front seat actuator and the rear seating area actuator for a single ignition cycle; disabling the eligibility of a specific passenger to initiate a request for the vehicle slowdown procedure; and disabling the eligibility of any passenger that is less than a pre-set age.
16. The system of claim 15, wherein the front seat actuator is at least one of a mechanical actuator, and an icon presented on one of the driver HMI or a front seat passenger HMI.
17. The system of claim 16, wherein the rear seating area actuator is at least one of a mechanical actuator, and an icon presented on a rear seat passenger HMI.
18. The system of claim 17, wherein each of the driver HMI, the front seat passenger HMI and the rear seat passenger HMI includes a microphone adapted to receive verbal input to the system controller.
19. The system of claim 11, wherein when initiating, via the vehicle control module, the automated slowdown procedure, the system controller is further adapted to: actuate automated driving algorithms within the vehicle control module; disable driver control of the vehicle; and cause, with the vehicle control module, the vehicle to slow down, maneuver off active lanes of a roadway on which the vehicle is traveling and stop.
20. A vehicle having a system for initiating an automated slowdown procedure within a vehicle, comprising: a system controller in communication with a driver human machine interface (HMI), wherein, the system controller is adapted to receive, via the driver HMI, preferences from a driver related to initiation of the automated slowdown procedure; an occupant monitoring system in communication with the system controller, wherein, the system controller is adapted to detect, with the occupant monitoring system, the presence and location of passengers, in addition to the driver, within the vehicle, identify, with the occupant monitoring system, at least one passenger located in a rear seating area of the vehicle, and determine, based on preferences from the driver, eligibility of the at least one passenger located in the rear seating area of the vehicle to request initiation of the automated slowdown procedure; a front seat actuator adapted to allow a request for initiation of the automated slowdown procedure to be made by at least one of the driver of the vehicle and a front seat passenger, and a rear seating area actuator adapted to allow a request for initiation of the automated slowdown procedure to be made by the at least one passenger located in the rear seating area, the system controller adapted to detect a request for initiation of the automated slowdown procedure by at least one of the front seat actuator and the rear seating actuator; the system controller further adapted to: prompt, via the driver HMI, the driver of the vehicle for verification to initiate the automated slowdown procedure; engage an automated driving mode; initiate, via a vehicle control module, the automated slowdown procedure when either one of: the driver provides, via the driver HMI, verification to initiate the automated slowdown procedure; or a response from the driver is not received within a pre-determined time; prompt, via the driver HMI, the driver of the vehicle with an option to selectively over-ride the automated slowdown procedure, and one of: continue with the automated slowdown procedure when the driver of the vehicle does not selectively over-ride, via the driver HMI, the automated slowdown procedure; or return control of the vehicle to the driver of the vehicle when the driver of the vehicle selectively over-rides, via the driver HMI, the automated slowdown procedure.